

# Shan Yang

Research Scientist — Multimodal RL & Reasoning Post-Training

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## PROFESSIONAL SUMMARY

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- Research scientist focused on **multimodal RL post-training** and visual-language reasoning, with 8+ years across foundation models at Adobe, Amazon, and Google Research
- **Sole-author arXiv 2026** on RL post-training for visual reasoning (**Physics-R1**: GSPO+DAPO, **+18.3 pp** on the held-out olympiad benchmark over Qwen3-VL-8B base); ICML 2026 co-author on test-time reasoning (**+35.8%** Math-500)
- Open-source releases at [huggingface.co/shanyangmie](https://huggingface.co/shanyangmie): PHYSCORP-A (6.4K audited multimodal physics corpus) and PHYSOLYM-A (500-problem held-out olympiad eval, 99.8% novel-source); prior first/co-first author at NeurIPS (Attention Bottlenecks) and ICCV (AI Choreographer)

## TECHNICAL SKILLS

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**RL post-training:** GRPO, GSPO, DAPO, DPO, RLHF, verl, MM-Eureka, reward shaping, LLM-as-judge protocols, audit-aware curricula

**Multimodal foundation models:** vision-language transformers, autoregressive video / diffusion, test-time training, attention-based fusion

**Infra:** PyTorch, FSDP / DeepSpeed, vLLM, distributed training at 10B+ params, HuggingFace, Croissant 1.0 + RAI dataset packaging

## SELECTED PUBLICATIONS

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### Physics-R1: An Audited Olympiad Corpus and Recipe for Visual Physics Reasoning

*S Yang*

arXiv 2026 (2605.14040) — project / code / data

*Sole-author. Three-stage corpus audit (Jaccard  $\rightarrow$  mxbai  $\rightarrow$  Haiku-4.5 judge), 500-problem novel-source held-out olympiad benchmark, and GSPO+DAPO recipe trained at  $\sim 360$  H200-GPU-hours over 3 seeds on Qwen3-VL-8B; +18.3 pp on PhysOlym-A, +15.7 pp on PhysReason. Method follow-up in preparation.*

### Beyond Test-Time Training: Learning to Reason via Hardware-Efficient Optimal Control

*P Wang, S Yang, X Wang, T Xiao, X Liu, C Yu, Y Lou, P Li, Z Wang, MC Lin, R Vidal*

ICML 2026

*Linear recurrent architecture leveraging optimal control theory for LLM test-time reasoning (+35.8% Math-500, +10% AIME25).*

### ViLA: Efficient Video-Language Alignment for Video Question Answering

*X Wang, J Liang, CK Wang, K Deng, Y Lou, MC Lin, S Yang*

ECCV 2024

### ICAR: Image-based Complementary Auto Reasoning

*X Wang, A Liang, J Liang, M Lin, Y Lou, S Yang*

AAAI 2024

### Attention Bottlenecks for Multimodal Fusion

*A Nagrani, S Yang, A Arnab, A Jansen, C Schmid, C Sun*

NeurIPS 2021 — **800+** citations

### AI Choreographer: Music Conditioned 3D Dance Generation with AIST++

*R Li\*, S Yang\*, DA Ross, A Kanazawa*

ICCV 2021 — **700+** citations

### Physics-Inspired Garment Recovery from a Single-View Image

*S Yang, Z Pan, T Amert, K Wang, L Yu, T Berg, MC Lin*

ACM ToG 2018 — **100+** citations

## Modeling Context in Referring Expressions

*L Yu, P Poirson, **S Yang**, A Berg, T Berg*

ECCV 2016 — **1600+** citations

## RESEARCH IN PROGRESS

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**Reasoning-method paper (Physics-R1 follow-up):** Novel multimodal RL post-training method extending the GSPO+DAPO recipe with judge-robust correctness signals and audit-aware curricula; targeting ICLR 2027.

## ENGINEERING & RESEARCH EXPERIENCE

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### Adobe Foundry

Staff Applied Scientist

San Jose, CA

March 2026 – Present

- Post-training research for Adobe Foundry’s generative AI foundation models powering Firefly (SFT, RLHF/DPO, LoRA, distillation, test-time training)

### Amazon (Prime Video & Search)

Senior Applied Scientist Tech Lead

Palo Alto & Sunnyvale, CA

September 2021 – March 2026

- **Prime Video (2025–2026):** Led GenAI Live Action Studio — autoregressive diffusion and training infrastructure for **10B+** **parameter** video generation; advanced controllable generation via test-time training, teacher-diffusion-forcing, and RL post-training
- **Search (2021–2025):** Pioneered multimodal fusion architecture (video / text / image / structured) for Amazon Video Search, deployed at scale; co-authored ViLA (ECCV 2024) and ICAR (AAAI 2024); led Schema-of-Thought research on multimodal LLM robustness (**+28%** hallucination reduction)

### Google Research

Senior Research Software Engineer

Mountain View, CA

June 2017 – September 2021

- First/co-first author publications at ICCV (AI Choreographer + AIST++ dataset), NeurIPS (Attention Bottlenecks), and ECCV; drove research agenda on music-conditioned 3D motion synthesis and multimodal generative modeling
- Owned 2D/3D pose reconstruction models deployed on GCP at scale

## EDUCATION

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### University of North Carolina at Chapel Hill

Ph.D., Computer Science

Chapel Hill, NC

2012 – 2017

### Shanghai Jiao Tong University

B.E., Software Engineering

Shanghai, China

2008 – 2012